Use Case: Grid Congestion Prices

Hackathon kit for student teams · Time box: about 5 hours

The question

Where and when do congestion-related electricity price impacts occur, and how do those impacts compare across locations?

What's in this kit

File	What it's for
Student_Guide.docx	The full brief: key ideas, levels, data, 5-hour plan, roles and judging rubric. Read this first.
Data_Dictionary.docx	Every column in every data file explained, plus how the data was built
data/Congestion_monthly_by_zone_2023_2025.csv	396 rows (11 zones × 36 months), with average LBMP and congestion cost. Start here
data/Congestion_Daily_by_zone_2023_2025.csv	1,096 rows. Daily average congestion cost for all 11 zones
data/Congestion_Hourly_by_Month_2023_2025.csv	864 rows. Average congestion cost by hour of day for each month
data/Congestion_Raw_2023_2025.csv	26,276 rows. Hourly congestion cost for all 11 zones
data/Mosesouth_flow_2023_2025_raw.csv	26,296 rows. Hourly power flow across the Moses South interface
data/Mosesouth_flow_2023_2025_dailyavg.csv	1,096 rows. Daily average Moses South flow
data/zones_reference.csv	11 rows. Zone names, regions and map coordinates

Pick a level

Level	Build	Deliverable
Minimum	Congestion price comparison	Identify the relationship between LBMP and congestion costs in a chart; find the three zones with the highest average LBMP; find the three zones with the heaviest negative congestion costs
Intermediate	Congestion pattern analyzer	Map time-series congestion costs for the top 3 zones in a chart; show findings on

Level	Build	Deliverable
		seasonal, monthly, hourly and weekend vs weekday variation
Advanced	Congestion mapping and connect with Smart Path flow (Moses South)	Trend analysis of Moses South flow from 2023 to 2025; explore the correlation between Moses South flow and congestion cost for zones D, F and J

Use any tools you like. We provide the data; your team decides how to analyze and present it (spreadsheets, Python, R, Power BI, Tableau, mapping tools or anything else).

Data source

NYISO real-time zonal LBMP (5-minute prices) and NYISO interface flow data for Moses South (5-minute flows), January 1 2023 to December 31 2025, summarized to hourly, daily and monthly values. Prices are in $/MWh. In these files a **negative** congestion value means congestion **raised** the price in that zone, because LBMP = energy + losses − congestion.